

Mixed-field tangent and tangent-only numerical perturbation

	u_x	u_y	p	S
u_x	$J_{u_x u_x}$	$J_{u_x u_y}$	$J_{u_x p}$	$J_{u_x S}$
u_y	$J_{u_y u_x}$	$J_{u_y u_y}$	$J_{u_y p}$	$J_{u_y S}$
p	$J_{p u_x}$	$J_{p u_y}$	$J_{p p}$	$J_{p S}$
S	$J_{S u_x}$	$J_{S u_y}$	$J_{S p}$	$J_{S S}$

u_x : displacement [m]

u_y : displacement [m]

p : pressure [Pa]

$$J_{\text{returned}} = J_{\text{phys}} + \Delta J_{\text{num}}$$

Red diagonal markers

tangent-only perturbation overlay

One scalar coefficient does not

define one scale across mixed fields.

Conceptual anatomy only; no UEL revalidation claim.
 S : saturation [-]